



Pharm*Achieve*

COMMUNITY AND HOSPITAL ACQUIRED PNEUMONIA

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

Abx	Antibiotics
ACE	Angiotensin II Converting Enzyme
Amox-Clav	Amoxicillin- Clavulanic Acid
ARDS	Acute Respiratory Distress Syndrome
β-lactam	Beta Lactam
BP	Blood Pressure
CAP	Community Acquired Pneumonia
CHF	Congestive Heart Failure
CNS	Central Nervous System
COPD	Chronic Obstructive Pulmonary Disorder
COVID	Coronavirus Disease
CT	Computed Tomography
DRSP	Drug Resistant S. pneumoniae
G	Gram
Gen	Generation
HAP	Hospital Acquired Pneumonia
HIV	Human Immunodeficiency Virus
ICU	Intensive Care Unit

IV	Intravenous
MDR	Multi-Drug Resistance
MRI	Magnetic Resonance Imaging
MRSA	Methicillin Resistant S. Aureus
MSSA	Methicillin Susceptible S. aureus
NG	Nasogastric
NSAIDs	Non-steroidal Anti-inflammatory Drugs
OG	Orogastric
PE	Pulmonary Embolism
PEG	Percutaneous Endoscopic Gastrostomy
Pip-Taz	Piperacillin-Tazobactam
PO	Oral
PSI	Pneumonia Severity Index
SOB	Shortness of Breath
Spp	Species
TID	Three Times Daily
TMP/SMX	Trimethoprim/Sulfamethoxazole
VAP	Ventilation Associated Pneumonia

WHAT IS PNEUMONIA?

Pneumonia is an acute infection of the pulmonary parenchyma

Community-Acquired Pneumonia (CAP)

- Pneumonia in a patient **not hospitalized** or hospitalized for less than 48 hours

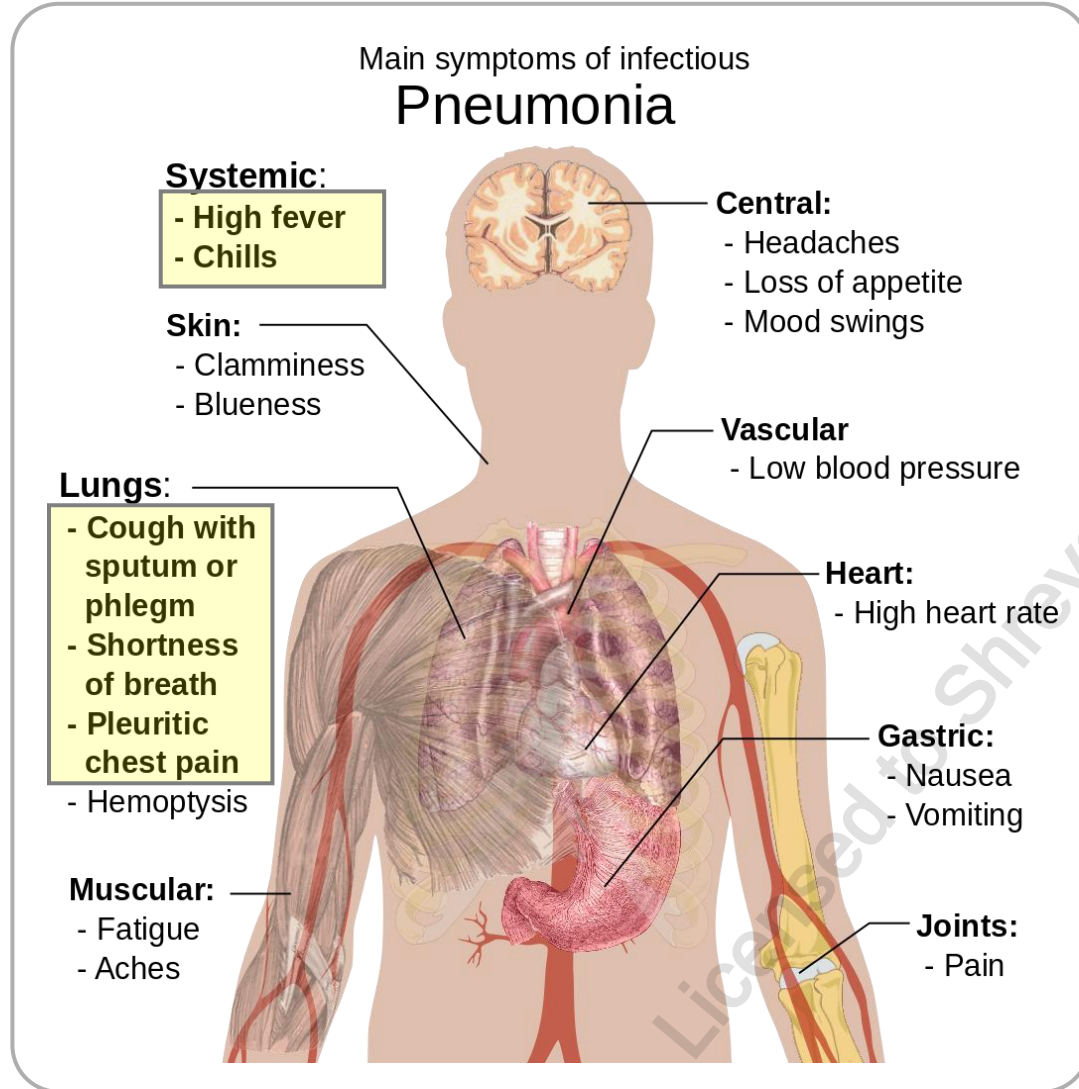
Hospital-Acquired Pneumonia (HAP)

- Defined as pneumonia occurring **after ≥ 48 hours of hospitalization** (not associated with mechanical ventilation)

Ventilator-Associated Pneumonia (VAP)

- Arises **48 hours after endotracheal intubation**

CLINICAL PRESENTATION AND PATHOPHYSIOLOGY



- Pathogens must gain access by one of four methods:
 - Aspiration of bacteria
 - Inhalation of aerosolized droplets containing pathogens
 - Hematogenous spread
 - Direct inoculation
- **Complications**
 - Bacteremia
 - Sepsis/septic shock
 - Empyema/parapneumonic effusion
 - Lung abscesses

ETIOLOGY AND DIAGNOSTIC TESTS

Patient Type	Common CAP Pathogens
Outpatient	Respiratory viruses (including influenza, COVID) <i>S. pneumoniae</i> <i>M. catarrhalis</i> <i>C. pneumoniae</i> <i>M. pneumoniae</i> <i>S. Aureus</i> (rare but severe) <i>H. Influenzae</i>
Inpatient (Non-ICU)	Outpatient pathogens, +/- gram negative bacilli
Inpatient (ICU)	<i>S. pneumoniae</i> <i>S. aureus</i> (including MRSA) <i>H. influenzae</i> <i>Legionella spp.</i> <i>Enterobacteriales</i>

Patient Type	Common HAP and VAP Pathogens
VAP and HAP	<i>S. aureus</i> (including MRSA) <i>P. aeruginosa</i> Enteric gram-negative bacilli <i>S. pneumoniae</i> (much less common)

Lab findings and diagnostic tests

1. **X-ray:** able to identify infiltrates (**GOLD STANDARD**)
2. **CT:** provide more detailed images
3. **MRI:** less commonly used for pneumonia diagnosis
4. **Culture and sensitivity:** sample of sputum and blood may identify pathogen (generally for severe CAP, HAP and VAP)

SEVERITY OF COMMUNITY ACQUIRED PNEUMONIA IN ADULTS

The decision to admit patients to the hospital is based on their signs and symptoms, as well as the likelihood of morbidity and mortality (determined by the PSI or CURB-65 scores)

Pneumonia Severity Index (PSI)	CURB-65 Score
- Looks at demographics, comorbid illnesses, physical examinations, vital signs and lab values - Predicts the 30-day mortality using a score (higher score indicates severity) - Class I (score <51) (0.1%) and II (51-70) (0.6%): outpatient treatment - Class III (score 71-90) (2.8%): brief hospitalization - Class IV (score 91-130) (8.2%): inpatient - Class V (score >130) (29.2%): ICU - Admit patients with PSI >71 to hospital	- Benefit: easier to use than PSI, with similar sensitivity/specificity 5 components, each worth 1 point - Confusion - Blood Urea Nitrogen >7 mmol/L - Respiratory rate ≥ 30 breaths/min - Blood pressure <90/60 mmHg - Age (≥65) - Score (mortality prediction) 0 = 0.6% 1 = 2.7% 2 = 6.8% 3 = 14% 4-5 = 57% - Admit patients with CURB-65 score ≥ 2 to hospital

RISK FACTORS FOR CAP

Developing pneumonia	Mortality	Both
• Decreased respiratory function (i.e., reduced mucociliary transport) • Smoking • Malnutrition • Previous CAP • Bronchiectasis • Patients with increased risk of aspiration: history of stroke, Parkinson's disease, decreased level of consciousness, alcohol use, benzodiazepines, antipsychotics, among others	• Increased delay before therapy (>8 hours) • Bacteremia • Severity (PSI/CURB)	• Very old or very young patients • Immunocompromised • Underlying diseases such as diabetes, congestive heart failure, COPD COPD increases risk for infection from <i>H. influenzae</i>, possibly <i>Pseudomonas</i>

GOALS OF THERAPY

- ✓ Eradicate the causal pathogen
- ✓ Eliminate signs and symptoms
- ✓ Prevent complications
- ✓ Prevent recurrence
- ✓ Minimize side effects of medication
- ✓ Minimize the development of resistant bacteria by using narrow spectrum antibiotics for the shortest effective duration

TREATMENT CONSIDERATIONS

General considerations

1. Use **empirical drug therapy** until culture results (if indicated) arrive
2. Consider **local resistance patterns**
3. Once the pathogen has been identified, change to **targeted therapy**
4. Symptoms **usually improve after 3-5 days:**

Treat the infection for approximately 5-7 days (i.e. patient has no fever for at least 72 hours and is clinically stable)

Special considerations

1. When starting empirical therapy, patients who have received an antibiotic in the past 3 months should be given an agent from a **different class of antibiotics** to minimize the risk of treatment failure due to antimicrobial resistance
2. Multi-Drug Resistance (MDR) should be considered if DRIP score ≥ 4

Major risk (2 pts each)	Minor risk (1 point each)
Antibiotics within 60 days	Hospitalization within 60 days
LTC resident	Pulmonary disease
NG/OG/PEG feeds	Poor functional status
Hx of MDR pneumonia in last year	H2RA/PPI in last 14 days

EMPIRIC THERAPY OF COMMUNITY-ACQUIRED PNEUMONIA

Patient type	Antibiotic options
Outpatient (no modifying factors)	- Amoxicillin (children and adults) - Doxycycline 100mg PO BID - Macrolide (azithromycin or clarithromycin) *depending on resistance* avoid if >25% local <i>S. pneumoniae</i> resistance Always consider local resistance patterns
Outpatient with modifying factors**	- Beta-lactam (amox/clav or 2nd gen cephalosporin) + macrolide/<i>doxycycline</i> - Respiratory fluoroquinolone
Inpatient Non-ICU	- Respiratory fluoroquinolone - Beta-lactam IV / PO (ceftriaxone, cefotaxime, or amox/clav) + macrolide CHILDREN ONLY: IV ampicillin OR PO amoxicillin
ICU	3rd generation cephalosporin IV + (fluoroquinolone IV OR azithromycin IV)
Inpatient/ICU <i>Pseudomonas</i> suspected	- Antipseudomonal beta-lactam (e.g., pip-taz, cefepime, ceftazidime, meropenem) + (ciprofloxacin OR levofloxacin) - Antipseudomonal beta-lactam + aminoglycoside + IV azithromycin

**Any respiratory, circulatory (not HTN), immunologic, rheumatologic, hepatic, renal or endocrinologic comorbidity

TARGETED THERAPY FOR COMMUNITY-ACQUIRED PNEUMONIA

Pathogens	Preferred treatment (pick one unless specified)
<i>S. pneumoniae</i> (consider resistance)	Penicillin, amoxicillin, cefotaxime, ceftriaxone
<i>H. influenzae</i>	Amoxicillin, amox/clav, 2 nd /3 rd gen. cephalosporins
<i>M. catarrhalis</i>	Amox/clav, 2 nd /3 rd gen. cephalosporins
<i>M. pneumoniae</i> / <i>C. pneumoniae</i>	Macrolide, doxycycline, respiratory FQ
<i>Legionella</i> spp.	Macrolide, doxycycline, respiratory FQ
MSSA	Cloxacillin (IV), cefazolin, cephalexin
MRSA	Vancomycin, linezolid (NOT daptomycin)

EMPIRIC THERAPY FOR HOSPITAL-ACQUIRED PNEUMONIA

	HAP	HAP	Severe HAP/all VAP	
	1 drug Stable (low Risk) patient/No MRSA - Pip/tazo - Cefepime - Levofloxacin - Meropenem - Imipenem	2 drug Stable (low risk) patient/Possible MRSA - Pip/tazo - Cefepime - Levofloxacin - Ciprofloxacin - Meropenem - Imipenem - Aztreonam - Vancomycin - Linezolid	3 drug Unstable/High Risk*/IV Abx Exposure - Pip/tazo - Cefepime - Meropenem - Imipenem - Aztreonam - Levofloxacin - Ciprofloxacin - Aminoglycoside - Vancomycin - Linezolid	Group A1 Group A2 Group B

MDR Pseudomonas Risk: > 10% ward level resistance to first line agents
MRSA risk: >20% ward level MRSA prevalence, or possible colonization

*High Risk: ventilated or in septic shock

TARGETED THERAPY FOR HOSPITAL-ACQUIRED PNEUMONIA

Pathogens	Preferred treatment (pick one unless specified)
<i>P. aeruginosa</i>	Ceftazidime, pip/tazo, cipro, meropenem/imipenem, etc.
Enterobacteriales (i.e. PEcK)	Amoxicillin/ampicillin, amox/clav, cefotaxime, ceftriaxone, fluoroquinolone, carbapenem
<i>E. cloacae</i> , <i>K. aerogenes</i> , <i>C. freundii</i>	Fluoroquinolone, TMP/SMX, carbapenem
MSSA	Cloxacillin (IV), cefazolin, cephalixin
MRSA	Vancomycin, linezolid (NOT daptomycin)

TAILORING AFTER EMPIRIC THERAPY

Consideration	CAP	HAP/VAP
Using respiratory cultures	Once pathogen/organism is identified → "de-escalate" to "streamlined" therapy Narrowest and safest agent available	
Total duration of antimicrobial therapy*	5-7 days	7 days *longer (10-14 days) in certain cases
Oral therapy?	OK to use/change to PO therapy when patient is stable (e.g. improving oxygenation, able to tolerate oral therapy, afebrile, hemodynamically stable)	

*Duration Of Abx Therapy for CAP: treat for minimum of 5 days. Before discontinuation, the patient must be afebrile for 48h-72h, should be clinically stable. In children, the usual duration is 7-10 days.

PREVENTION AND PROPHYLAXIS

1. Prevention

1. Childhood: vaccination with conjugate pneumococcal vaccination
2. Adult: vaccination with Pneu-C-20 or Pneu-C-21 pneumococcal vaccine
3. All ages: influenza vaccine, COVID vaccine, routine hand/respiratory measures (hand washing, covering coughs, etc.)

2. Prophylaxis: only indicated in patients with specific conditions (e.g. HIV, active malignancy, etc.) for prevention of specific pneumonias (see HIV lecture for full details)

1. *Pneumocystis jirovecii*:

1. First line: TMP/SMX 1 DS 3x/week or 1 SS daily
2. Alternate (one of):
 1. Dapsone 100mg PO daily
 2. Atovaquone 1500mg PO daily
 3. Pentamidine monthly

MONITORING PARAMETERS

Parameter	Considerations
Fever and chills	Should resolve in 24-48 hours
Difficulty breathing	Improve in 48 hours; resolve after 3-5 days but may be delayed
Sputum production	Reduced in 48 hours; resolved after 3-5 days (sometimes longer – 1-2 weeks)
Cough	Reduced in 48 hours; resolved after 3-5 days (sometimes longer – 1-2 weeks)
WBC	Should improve in 48 hours
Chest radiograph	Repeating chest X-ray is not helpful unless condition worsening Improvement seen over 3-5 days; typically takes $\geq 4-6$ weeks to clear completely



Review reasons for failure (non-responsive to antibiotic therapy): wrong antibiotic (resistance), wrong microorganism (consider microbes not initially suspected including TB, dimorphic fungi, yeasts), wrong diagnosis (atelectasis, CHF, PE, pneumonitis, ARDS, cancer), consider complications (drug fever, abscess)

TAKEAWAYS

- Pneumonia is a common, potentially serious infection of the lungs that can be caused by a variety of different microorganisms
- Treatment is dependent upon the severity of the infection, the known or likely microbiology and host factors, including the location of pneumonia acquisition (hospital or community).
- Empiric management with antimicrobials directed against *S. pneumoniae* and other common respiratory bacteria are the mainstay of therapy
- Combination therapy may be warranted in patients with community-acquired pneumonia with comorbidities and/or those who are admitted to hospital
- The more severe the pneumonia, the less room for error. Broader coverage for more sick patients, until a pathogen is isolated.
- Empiric therapy for CAP – generally, 5 to 7-day treatment:
 - **Outpatient:** amoxicillin OR doxycycline OR macrolide* (* less preferred → increased *S. pneumoniae* resistance)
 - **Outpatient (with risk factors):** [amox/clav OR 2nd gen cephalosporin + (macrolide OR doxycycline)] OR Respiratory fluoroquinolone (e.g. moxifloxacin, levofloxacin)
 - **Inpatient (Non-ICU):** [amox/clav OR 3rd gen cephalosporin] + macrolide] OR Respiratory fluoroquinolone (e.g. moxifloxacin, levofloxacin)

CASE 1

JT is a 62-year-old male who presents to the ER with a 3-day history of worsening fever, chills, shortness of breath, and productive cough. JT reports that he began experiencing shortness of breath over a week ago, while going for a walk in the cold. After a few days of not feeling well, his family doctor prescribed levofloxacin 750 mg PO once daily for 5 days, which JT did not fill. He began to experience chest pain with a productive cough and rust-coloured sputum over the past 3 days, as well as a fever. JT has a past medical history of hypertension, type 2 diabetes mellitus, and asthma. He currently takes metformin 1000 mg PO BID, lisinopril 10 mg PO once daily, Advair® Diskus (fluticasone propionate/salmeterol) 500/50 mcg – 1 inhalation PO BID, and acetaminophen 650 mg PO Q4-6H PRN for fever. He has no allergies to medications. He is a retired teacher living with his wife, and he has 2 adult children who no longer live with him.

Objective findings:

- BP 160/85 mmHg, P 120, RR 30 breaths/min, T= 39.5°C
- CXR: right lower lobe consolidation



CASE 1 - QUESTIONS

1. What would have been the preferred antimicrobial agent for JT as an outpatient?
 - a) Azithromycin + Ceftriaxone
 - b) Doxycycline
 - c) Amoxicillin/Clavulanate + Azithromycin
 - d) Amoxicillin

2. What is the preferred antimicrobial agent for JT as an inpatient with a CURB score of 2?
 - a) Piperacillin-tazobactam
 - b) Ceftriaxone
 - c) Meropenem
 - d) Ceftriaxone + azithromycin



CASE 1 - QUESTIONS

3. JT's sputum culture grows beta-lactamase negative *H. influenza* on day 3 of his hospital stay. He is tolerating oral intake. What is the most reasonable treatment option?
- a) Azithromycin
 - b) Doxycycline
 - c) Amoxicillin
 - d) Moxifloxacin
4. Assuming he has no fever at 48 hours after antimicrobial initiation, what is the appropriate total duration of antimicrobial therapy for JT?
- a) 3 days
 - b) 5 days
 - c) 7 days
 - d) 10 days



CASE 2

CH is a 60-year-old female who, during her hospitalization for a hysterectomy, developed shortness of breath and increased sputum production. CH was recently diagnosed with ovarian cancer and was admitted for a scheduled hysterectomy. Her procedure went well, but 7 days later, she developed shortness of breath with worsening cough and decreased LOC which required transfer to the ICU for intubation. She has a past medical history of ovarian cancer and long QT syndrome. She currently takes enoxaparin 40mg subcutaneously once daily and hydromorphone 1mg po q4h prn. She has no allergies to medications.

Objective findings:

- CXR: lower left lung consolidation
- Vitals: RR 45 breaths/min, BP 160/100 mmHg, HR 100 bpm, O2 sat 85%, T 38.5°C
- ECG: Tachycardia, QTc 520 msec
- MRSA negative on admission
- Ward MRSA/total Staph = 0.25



CASE 2 - QUESTIONS

1. What is the most likely diagnosis for CH from an infectious perspective?
 - a) CAP
 - b) HAP
 - c) VAP
 - d) Influenza

2. What pathogens should we consider covering for CH that would not be empirically covered for JT?
 - a) *Legionella*
 - b) Methicillin-sensitive *Staphylococcus aureus*
 - c) Gram positive bacilli
 - d) Methicillin-resistant *Staphylococcus aureus*



CASE 2 - QUESTIONS

3. What is the MOST appropriate empiric antimicrobial regimen for CH?
- a) Piperacillin/tazobactam + vancomycin
 - b) Piperacillin/tazobactam + aztreonam + gentamicin
 - c) Meropenem + Levofloxacin + Linezolid
 - d) Cefepime + Tobramycin + Vancomycin
4. What is the MOST likely treatment-related adverse effect associated with the antimicrobial therapy for CH?
- a) QT prolongation
 - b) *C.difficile*
 - c) Seizure
 - d) Hepatotoxicity



REFERENCES

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7. Howroyd F, Gill R, Thompson J, et al. Ventilator-associated pneumonia: mechanisms, an appraisal of current therapies and the role for inhaled antibiotics in prevention and treatment. *Respir Med*. 2025;247:108275. doi:10.1016/j.rmed.2025.108275.

CHANGE LOG

June 2025

- Grammar and spelling changes made throughout

August 2025

- Slide 6: Removed 'FYI' on lab findings + diagnostic tests

October 2025

- Slide 18: Elaborated on Takeaway slide
- Formatting changes, no changes made to content

February 2026

- Slide 2: Updated competencies
- Slide 4: Edited to greater than or equal to 48 hours for HAP
- Slide 6: Added rarity of *S. Aureus* in outpatient CAP
- Slide 16: Revised vaccine recommendations for adults and prophylaxis
- Slide 25: Updated references